5 Solutions to Power Question

1. Compute from the given definition:

a. $|7-2|+|-2-6|=13$.

b. There are four points that can found by moving from $(3,2)$ along vectors parallel to the coordinate axes. These points are $(8,2),(-2,2),(3,7)$, and $(3,-3)$. *One of these is sufficient to answer the question.*

c. The classic $3-4-5$ right triangle gives many more possibilities: $(3\pm 3, 2\pm 4)$ and $(3\pm 4, 2\pm 3)$. (One of these is sufficient to answer the question.)

2. **a**. $d_T(A,P)+d_T(P,B)=|1-0|+|3-4|+|4-1|+|0-3|=8=|4-0|+|0-4|=d_T(A,B)$.

b. Many solutions exist (see part c below), but $(2,4)$ is one such point.

c. Because $d_T(A,B)=8$, to move from A to B, find all points $P(x,y)$ such that $|x-0|+|y-4|+|4-x|+|0-y|=8$. If $0\le x\le 4$ and $0\le y\le 4$, the left hand side of the equation simplifies to $x+(4-y)+(4-x)+y$, which equals 8 for all such x and y. On the other hand, if $x<0$ and $0\le y\le 4$, then the left hand side of the equation simplifies to $4-2x+(4-y)+y=8-2x$, and for $x<0$, $8-2x>8$. Analogous arguments show that no points with $x>4$, or $y<0$, or $y>4$ satisfy the condition. Hence the solution set is the square $0\le x\le 4, 0\le y\le 4$.

Alternate solution: Let $P=(x,y)$, so that $d_T(A,P)+d_T(P,B)=|x-0|+|y-4|+|4-x|+|0-y|=|4-x|+|x-0|+|4-y|+|y-0|$. For any real number r, $|r|\ge r$, and equality holds if and only if $r\ge 0$. Applying this observation to each term in the above sum, $d_T(A,P)+d_T(P,B)\ge (4-x)+(x-0)+(4-y)+(y-0)=8=d_T(A,B)$, with equality if and only if each of $4-x$, $x-0$, $4-y$, $y-0$ is nonnegative. Hence the solution set is the square $0\le x\le 4$ and $0\le y\le 4$.

3. **a**. Let $A=(x_1,y_1)$ and let $B=(x_2,y_2)$. Then $d_T(A,B)=|x_2-x_1|+|y_2-y_1|$. On the other hand, $A'=(x_1+h,y_1+k)$ and $B'=(x_2+h,y_2+k)$, yielding $d_T(A',B')=|(x_2+h)-(x_1+h)|+|(y_2+k)-(y_1+k)|=|x_2-x_1|+|y_2-y_1|=d_T(A,B)$. The same argument would apply to the taxi-lengths of $\overline{BC}$ and $\overline{AC}$. Therefore the taxi-lengths of the sides are preserved under translation.

b. Let $A=(x_1,y_1)$ and let $B=(x_2,y_2)$. The reflection takes $(x,y)=(x,k+(y-k))$ to $(x,k-(y-k))=(x,2k-y)$. Hence $A'=(x_1,2k-y_1)$ and $B'=(x_2,2k-y_2)$. Thus $d_T(A',B')=|x_2-x_1|+|(2k-y_2)-(2k-y_1)|=|x_2-x_1|+|y_1-y_2|=d_T(A,B)$. Analogous arguments apply to the taxi-lengths of $\overline{BC}$ and $\overline{AC}$, so the taxi-lengths of the sides are preserved under reflection in a horizontal line. (Note: It would be straightforward to show the same result for reflection in a vertical line. What might be difficult about reflecting in an oblique line?)

c. $O'=(0,0)$, $R'=(4\sqrt{2},4\sqrt{2})$, $D'=(-2\sqrt{2},2\sqrt{2})$. Notice that $d_T(O,D)=4$, $d_T(D,R)=12$, and $d_T(O,R)=8$, but $d_T(O',R')=8\sqrt{2}$. Therefore $\triangle ORD\ncong_T\triangle O'R'D'$.

4. **a**. Let θ be the angle between the positive x-axis and the ray $\overrightarrow{OA}$. Then $\tan\theta=\frac{1}{2}$. The ray $\overrightarrow{OB}$ makes an angle of 2θ with the positive x-axis, so using the double-angle identity from trigonometry yields $\tan 2\theta=\dfrac{2\tan\theta}{1-\tan^2\theta}=\dfrac{1}{1-\left(\frac{1}{2}\right)^2}=\dfrac{4}{3}$. Hence B lies on the line $y=\frac{4}{3}x$. Because m and n can have no prime factors in common, $m=3$ and $n=4$. The desired point is $(3,4)$.

b. The rays $\overrightarrow{OA}$ and $\overrightarrow{OB}$ intersect the line $x+y=1$ at $\left(\frac{2}{3},\frac{1}{3}\right)$ and $\left(\frac{3}{7},\frac{4}{7}\right)$ respectively. Note that $d_T\left(P,\left(\frac{2}{3},\frac{1}{3}\right)\right)=|1-\frac{2}{3}|+|0-\frac{1}{3}|=\frac{2}{3}$. However, $d_T\left(\left(\frac{2}{3},\frac{1}{3}\right),\left(\frac{3}{7},\frac{4}{7}\right)\right)=|\frac{3}{7}-\frac{2}{3}|+|\frac{4}{7}-\frac{1}{3}|=\frac{10}{21}$. Therefore central angles with the same Euclidean angle measure can intercept arcs of different taxi-lengths on the unit taxi-circle.

5. **a**. The taxi-circle is the set of all points (x,y) such that $|x|+|y|=17$. This equation describes a square centered at the origin with vertices $(\pm 17,0)$ and $(0,\pm 17)$; the sides of the square lie on the lines $x+y=17$, $x+y=-17$, $x-y=17$, and $x-y=-17$.

b. The semi-taxi-circumference of this circle is 68, because $d_T((17,0),(0,17))=d_T((0,17),(-17,0))=34$. (Note that the distance along the arc is, as usual, longer than the straight-line taxi-distance $d_T((17,0),(-17,0))$, which is simply 34.) The radius of the circle is 17. So $\pi_T=68/17=4$.

c. In general, the taxi-circle of radius r centered at the origin is given by the equation $|x| + |y| = r$, and one semicircle has endpoints $(r, 0)$ and $(-r, 0)$. Hence the arc length is simply $d_T((r, 0), (0, r)) + d_T((0, r), (-r, 0)) = 2r + 2r = 4r$. Therefore $\pi_T = 4r/r = 4$. (As an interesting exercise, you might try to show that this result holds for *any* taxi-semicircle with endpoints (x, y) and $(-x, -y)$.)

d. These points are both found on a taxi-circle of radius 4.

$$\mathrm{m}_T\angle COB = \frac{\text{taxi-length of the arc from } C \text{ to } B \text{ on taxi-circle } O}{d_T(O, C)} = \frac{8}{4} = 2 \text{ taxi-radians.}$$

6. **a**. First note that $O' = O$ because O is the center of rotation. To find A', note that $d_T(O, A) = 8$, so the arc from A to A' on the circle centered at O passing through A must also have length 8, and if A' is in the first quadrant, then it lies on the line $x + y = 8$. These conditions are satisfied by the point $(4, 4)$. Similarly, R' must lie on the line $x + y = 4$ and $d_T(R, R') = 4$. Assuming that $0 < x < 3$ and $1 < y < 4$ yields $d_T(R, R') = (3 - x) + (y - 1) = y - x + 2$. Hence the desired point satisfies both $x + y = 4$ and $y - x + 2 = 4$, yielding $x = 1, y = 3$ as the unique solution. Thus $R' = (1, 3)$.

Alternate solution: First note that $O' = O$ because O is the center of rotation. Before considering the other points, let $P = (x, y)$ be any point in the first quadrant and let $0 \le r \le x$. Then $Q = (x - r, y + r)$ is also in the first quadrant, Q and P are on the same taxi-circle centered at O, and $d_T(P, Q) = 2r$. Apply this observation to $A = (8, 0)$ with $r = \frac{1}{2}d_T(O, A) = 4$ to find $A' = (8-4, 0+4) = (4, 4)$. Similarly, take $P = B = (3, 1)$ and $r = \frac{1}{2}d_T(O, B) = 2$ to find $B' = (3 - 2, 1 + 2) = (1, 3)$.

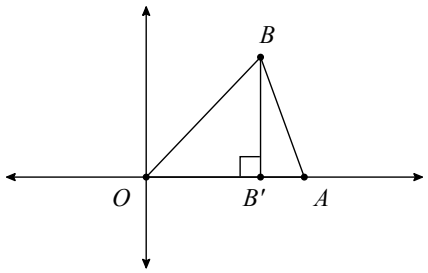
b. By construction, $d_T(O, A) = d_T(O, A')$ and $d_T(O, B) = d_T(O, B')$. However, $d_T(A, B) = |3-8|+|1-0| = 6$ while $d_T(A', B') = |1 - 4| + |3 - 4| = 4$. Hence the two triangles are *not* measure-congruent.

7. **a**. There are many examples. One such example is $\triangle CAB$ with vertices $C(-1, 1)$, $A(0, 0)$, and $B(1, 1)$. Then $d_T(C, A) = d_T(M, O) = 2$, $d_T(A, B) = d_T(O, D) = 2$, and $\mathrm{m}_T\angle A \cong \mathrm{m}_T\angle O = \pi_T/2$, but $\triangle MOD \ncong_T \triangle CAB$ because $d_T(C, B) = 2 \neq d_T(M, D)$.

b. There are many examples. One such example is $\triangle PIG$ with vertices $P(-1, 1)$, $I(0, 0)$, and $G(2, 0)$. Then $d_T(P, I) = d_T(M, O) = 2$, $d_T(I, G) = d_T(O, D) = 2$, and $d_T(P, G) = d_T(M, D) = 4$, but $\triangle MOD \ncong_T \triangle PIG$ because $\mathrm{m}_T\angle PIG > \mathrm{m}_T\angle MOD$.

8. **a**. To use the formula, note that $d_E(O, A) = 3$, $d_E(O, B) = 5$, $d_T(O, A) = 3$, $d_T(O, B) = 7$, $\cos\angle AOB = \frac{3}{5}$, and $\sin\angle AOB = \frac{4}{5}$. Then $\cos_T\angle AOB = \frac{3\cdot 5}{3\cdot 7} \cdot \frac{3}{5} = \frac{3}{7}$. (Similarly, $\sin_T\angle AOB = \frac{3\cdot 5}{3\cdot 7} \cdot \frac{4}{5} = \frac{4}{7}$.)

b. Consider the diagram below.



Because $\overrightarrow{OA}$ is parallel to the x-axis, $d_E(O, A) = d_T(O, A)$. Then

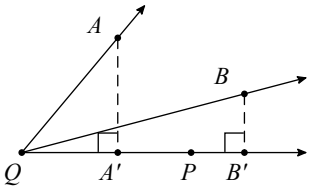
$$\cos_T\angle AOB = \frac{d_E(O, A) \cdot d_E(O, B)}{d_T(O, A) \cdot d_T(O, B)} \cos\angle AOB = \frac{d_E(O, B)}{d_T(O, B)} \cos\angle AOB.$$

Let B' be the foot of the perpendicular from B to $\overleftrightarrow{OA}$, so that $d_T(O, B) = d_E(O, B') + d_E(B', B)$. Then

$$\cos_T\angle AOB = \frac{d_E(O, B)}{d_E(O, B') + d_E(B', B)} \cos\angle AOB = \frac{1}{\frac{d_E(O,B')}{d_E(O,B)} + \frac{d_E(B,B')}{d_E(O,B)}} \cos\angle AOB.$$

Substituting $|\cos\angle AOB|$ for $\frac{d_E(O,B')}{d_E(O,B)}$ and $|\sin\angle AOB|$ for $\frac{d_E(B,B')}{d_E(O,B)}$ yields the desired result. (The reader may wish to show that, under these conditions, $\cos_T\angle AOB + \sin_T\angle AOB = 1$.)

c. Let the feet of the altitudes from A and B to $\overleftrightarrow{PQ}$ be A' and B', respectively, as shown in the diagram below.



Then $d_T(Q,A) = d_E(Q,A') + d_E(A',A)$ and $d_T(Q,B) = d_E(Q,B') + d_E(B',B)$. Using the definition yields

$$\begin{aligned}
\cos_T\angle AQB &= \frac{d_E(Q,A)\cdot d_E(Q,B)}{d_T(Q,A)\cdot d_T(Q,B)}\cos\angle AQB \\
&= \frac{1}{\left(\dfrac{d_E(Q,A')+d_E(A,A')}{d_E(Q,A)}\right)\cdot\left(\dfrac{d_E(Q,B')+d_E(B,B')}{d_E(Q,B)}\right)}\cos\angle AQB \\
&= \frac{1}{\left(\dfrac{d_E(Q,A')}{d_E(Q,A)}+\dfrac{d_E(A,A')}{d_E(Q,A)}\right)\cdot\left(\dfrac{d_E(Q,B')}{d_E(Q,B)}+\dfrac{d_E(B,B')}{d_E(Q,B)}\right)}\cos\angle AQB.
\end{aligned}$$

On the other hand, $\frac{d_E(Q,A')}{d_E(Q,A)} = |\cos\angle AQP|$, $\frac{d_E(A',A)}{d_E(Q,A)} = |\sin\angle AQP|$, and similarly $\frac{d_E(Q,B')}{d_E(Q,B)} = |\cos\angle BQP|$, $\frac{d_E(B',B)}{d_E(Q,B)} = |\sin\angle BQP|$. Hence the right side can be rewritten as

$$\frac{\cos\angle AQB}{(|\cos\angle AQP| + |\sin\angle AQP|)(|\cos\angle BQP| + |\sin\angle BQP|)},$$

which is the desired result.

Alternate solution: Let the feet of the altitudes from A and B to $\overleftrightarrow{PQ}$ be A' and B', respectively. Then

$$\begin{aligned}
d_T(Q,A) &= d_E(Q,A') + d_E(A',A) \\
&= d_E(Q,A)\cdot|\cos\angle AQP| + d_E(Q,A)\cdot|\sin\angle AQP| \\
&= d_E(Q,A)\cdot\big(|\cos\angle AQP| + |\sin\angle AQP|\big),
\end{aligned}$$

so $d_E(Q,A)/d_T(Q,A) = \big(|\cos\angle AQP| + |\sin\angle AQP|\big)^{-1}$. The analogous formula for $d_E(Q,B)/d_T(Q,B)$ is derived in the same way. Substituting these expressions into the definition of the taxi-cosine yields

$$\begin{aligned}
\cos_T\angle AQB &= \frac{d_E(Q,A)\cdot d_E(Q,B)}{d_T(Q,A)\cdot d_T(Q,B)}\cos\angle AQB \\
&= \frac{\cos\angle AQB}{(|\cos\angle AQP| + |\sin\angle AQP|)(|\cos\angle BQP| + |\sin\angle BQP|)},
\end{aligned}$$

which is the desired result.

9. For these three problems, the following lemma will be useful:

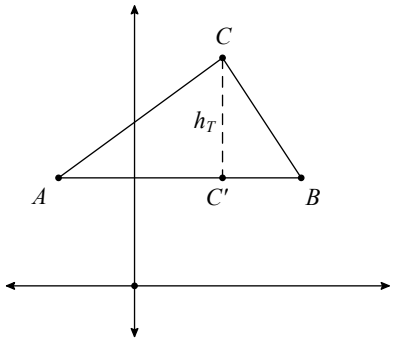
Lemma: Suppose that $\triangle PQR$ is a right triangle with right angle at Q and $\overline{PQ}$ parallel to the x-axis. Then $\cos_T\angle P = \frac{d_T(P,Q)}{d_T(P,R)}$.

Proof of Lemma: Because $\overline{PQ}$ is parallel to the x-axis, the result from 8b applies, and $\cos_T \angle P = \frac{\cos \angle P}{|\cos \angle P| + |\sin \angle P|}$. In this situation, $\cos \angle P = \frac{d_E(P,Q)}{d_E(P,R)}$ and $\sin \angle P = \frac{d_E(Q,R)}{d_E(P,R)}$, so substitute to obtain the following:

$$\begin{aligned} \cos_T \angle P &= \frac{\frac{d_E(P,Q)}{d_E(P,R)}}{\left|\frac{d_E(P,Q)}{d_E(P,R)}\right| + \left|\frac{d_E(Q,R)}{d_E(P,R)}\right|} \\ &= \frac{d_E(P,Q)}{|d_E(P,Q)| + |d_E(Q,R)|}. \end{aligned}$$

However, P and Q lie on the same horizontal line and Q and R lie on the same vertical line, so $d_E(P,Q) = d_T(P,Q)$ and $d_T(P,R) = d_E(P,Q) + d_E(Q,R)$, yielding the desired result. □

Consider the diagram below.



a. Because $d_T(A,C) = d_T(A,C') + d_T(C',C)$, it follows that $d_T(C,C') = d_T(A,C) - d_T(A,C')$ and analogously that $d_T(C,C') = d_T(B,C) - d_T(B,C')$. Then $d_T(B,C) = d_T(A,C) + d_T(A,B) - 2 \cdot d_T(A,C') = b_T + c_T - 2d_T(A,C')$. The lemma applies to angle A in $\triangle AC'C$, and $\cos_T \angle CAC' = \dfrac{d_T(A,C')}{d_T(A,C)}$ implies $d_T(A,C') = d_T(A,C) \cdot \cos_T \angle A$. Substituting this expression for $d_T(A,C')$ into the equation for $d_T(B,C)$ yields the first equation.

b. The second equation is derived similarly. As before, $d_T(A,C) = d_T(B,C) + d_T(A,B) - 2 \cdot d_T(A,C') = a_T + c_T - 2d_T(A,C')$. Again, apply the lemma to angle B in $\triangle BC'C$ to obtain $\cos_T \angle CBC' = \dfrac{d_T(B,C')}{d_T(B,C)}$, yielding $d_T(B,C') = d_T(B,C) \cdot \cos_T \angle B$. Substituting this result into the previous equation yields $d_T(A,C) = a_T + c_T - 2a_T \cos_T \angle B$.

c. Apply the definition of taxi-cosine to obtain $\cos_T \angle ACB = \dfrac{d_E(A,C) \cdot d_E(B,C)}{d_T(A,C) \cdot d_T(B,C)} \cos \angle ACB$. By the Law of Cosines applied to $\triangle ABC$, $\cos \angle C = \dfrac{(d_E(B,C))^2 + (d_E(A,C))^2 - (d_E(A,B))^2}{2 \cdot d_E(A,C) \cdot d_E(B,C)}$. Hence

$$\begin{aligned} \cos_T \angle ACB &= \frac{d_E(A,C) \cdot d_E(B,C)}{d_T(A,C) \cdot d_T(B,C)} \cdot \frac{(d_E(B,C))^2 + (d_E(A,C))^2 - (d_E(A,B))^2}{2 \cdot d_E(A,C) \cdot d_E(B,C)} \\ &= \frac{(d_E(B,C))^2 + (d_E(A,C))^2 - (d_E(A,B))^2}{2 \cdot d_T(A,C) \cdot d_T(B,C)}. \end{aligned}$$

Use the Pythagorean Theorem in $\triangle AC'C$ and $\triangle BC'C$ to obtain

$$\cos_T \angle ACB = \frac{(d_E(B,C'))^2 + (d_E(C,C'))^2 + (d_E(A,C'))^2 + (d_E(C,C'))^2 - (d_E(A,C') + d_E(B,C'))^2}{2 \cdot d_T(A,C) \cdot d_T(B,C)}.$$

By algebra, this equation is equivalent to $\cos_T \angle ACB = \dfrac{h_T^2 - p_T(c_T - p_T)}{a_T \cdot b_T}$, where $p_T = d_T(A,C')$.
Now, obtain expressions for h_T, p_T and $c_T - p_T$ by algebraic manipulation of previously-found equations

relating lengths of sides.

First, find an expression for p_T. Note that $d_T(A,C) = b_T = a_T + c_T - 2a_T \cos_T \angle B$, which implies $b_T = a_T + c_T - 2a_T \left(\dfrac{d_E(B,C) \cdot (c_T - p_T)}{a_T \cdot (c_T - p_T)} \cdot \dfrac{c_T - p_T}{d_E(B,C)} \right)$. After cancellation, $b_T = a_T + c_T - 2(c_T - p_T) \rightarrow b_T = a_T - c_T + 2p_T \rightarrow p_T = \dfrac{b_T + c_T - a_T}{2}$.

A similar manipulation to the equation $b_T = a_T + c_T - 2(c_T - p_T)$ yields $c_T - p_T = \dfrac{a_T + c_T - b_T}{2}$.

Now, notice that $h_T = a_T - (c_T - p_T) = a_T - \dfrac{a_T + c_T - b_T}{2}$, which simplifies to $\dfrac{2a_T}{2} - \dfrac{a_T + c_T - b_T}{2} = \dfrac{a_T - c_T + b_T}{2}$, as needed. Substitution yields

$$\cos_T \angle ACB = \frac{\left(\dfrac{b_T + c_T - a_T}{2}\right)^2 - \left(\dfrac{b_T + c_T - a_T}{2}\right)\left(\dfrac{a_T + c_T - b_T}{2}\right)}{a_T \cdot b_T},$$

which simplifies to $\cos_T \angle ACB = \dfrac{a_T^2 + b_T^2 - c_T(a_T + b_T)}{2 \cdot a_T \cdot b_T}$.

This equation can be algebraically manipulated to obtain the desired result.

Alternate solution: To simplify the computations, introduce notation for various Euclidean lengths. Let

$$\begin{gathered} a = d_E(B,C), \qquad b = d_E(A,C), \qquad c = d_E(A,B) = c_T; \\ p = d_E(A,C') = d_T(A,C') = b\cos A; \\ q = d_E(B,C') = d_T(B,C') = a\cos B. \end{gathered}$$

Then

$$a_T = q + h_T, \qquad b_T = p + h_T, \qquad c_T = p + q.$$

a. From the definition of the taxi-cosine, $b_T \cos_T A = b_T \cdot \frac{b \cdot c}{b_T \cdot c_T} \cos A = b\cos A = p$. Therefore $b_T + c_T - 2b_T \cos_T A = (p + h_T) + (p + q) - 2p = q + h_T = a_T$.

b. Similarly, $a_T \cos_T B = q$, so $a_T + c_T - 2a_T \cos_T B = (q + h_T) + (p + q) - 2q = b_T$.

c. The Euclidean radian measures of the angles add up to π, so $\cos C = \cos(\pi - A - B) = -\cos(A + B) = \sin A \sin B - \cos A \cos B$. Substitute into the definition of the taxi-cosine and multiply by $a_T b_T$:

$$\begin{aligned} a_T b_T \cos_T C &= a_T b_T \frac{ab}{a_T b_T}(\sin A \sin B - \cos A \cos B) \\ &= (b\sin A)(a\sin B) - (b\cos A)(a\cos B) \\ &= h_T^2 - pq \\ &= h_T^2 - (b_T - h_T)(a_T - h_T) \\ &= (a_T + b_T)h_T - a_T b_T. \end{aligned}$$

Therefore $a_T^2 + b_T^2 - 2a_T b_T \cos_T C = a_T^2 + b_T^2 - 2(a_T + b_T)h_T + 2a_T b_T = (a_T + b_T)(a_T + b_T - 2h_T) = (a_T + b_T)c_T$. Solving for c_T gives the desired result.

10. In general, the area of a triangle ABC is equal to half of the area of the parallelogram determined by any two sides of triangle ABC. The definition of taxi-sine, namely that $\sin_T \angle AOB = \dfrac{d_E(O,A) \cdot d_E(O,B)}{d_T(O,A) \cdot d_T(O,B)} \sin \angle AOB$, implies that $d_T(O,A) \cdot d_T(O,B) \cdot \sin_T \angle AOB = d_E(O,A) \cdot d_E(O,B) \cdot \sin \angle AOB$. The latter expression is equal to $|\overrightarrow{OA} \times \overrightarrow{OB}|_T$. Thus the cross-product of two vectors can be interpreted in a taxicab space as it is in a

Euclidean space!
Now, express the area of the triangle ABC in three different ways:

$$[ABC] = \frac{|\overrightarrow{AB} \times \overrightarrow{AC}|_T}{2} = \frac{|\overrightarrow{AB} \times \overrightarrow{BC}|_T}{2} = \frac{|\overrightarrow{AC} \times \overrightarrow{BC}|_T}{2}.$$

Notice also that $|\overrightarrow{AB} \times \overrightarrow{AC}|_T = c_T \cdot b_T \cdot \sin_T A$ and $|\overrightarrow{AB} \times \overrightarrow{BC}|_T = c_T \cdot a_T \cdot \sin_T B$ and $|\overrightarrow{AC} \times \overrightarrow{BC}|_T = b_T \cdot a_T \cdot \sin_T C$. By substitution and multiplication by 2, $c_T \cdot b_T \cdot \sin_T A = c_T \cdot a_T \cdot \sin_T B = b_T \cdot a_T \cdot \sin_T C$. Dividing by the nonzero quantity $a_T b_T c_T$ and taking reciprocals leads to the desired result.

Alternate solution: As in the solution to problem 9, let a, b, and c denote the Euclidean lengths of the sides. With this notation, the definition of the taxi-sine gives

$$\sin_T C = \frac{ab}{a_T b_T} \sin C = \frac{ab \sin C}{a_T b_T}, \quad \text{thus} \quad \frac{c_T}{\sin_T C} = \frac{a_T b_T c_T}{abc} \frac{c}{\sin C}.$$

Hence the Taxicab Law of Sines follows from the Euclidean version, multiplying by $\dfrac{a_T b_T c_T}{abc}$.

Note: Taxicab geometry is an active area of mathematical research, so many important questions remain unanswered. Students who are interested in taxicab geometry might find the following sources helpful; both were inspirational in the creation of this Power Question.

Eugene Krause, *Taxicab Geometry: An Adventure in Non-Euclidean Geometry*, Dover, 1975.

Ayşe Bayar, Süheyla Ekmekçi and Münevver Özcan, *On Trigonometric Functions and Cosine and Sine Rules in Taxicab Plane*, International Electronic Journal of Geometry, **2** (2009), 17-24. Retrieved from http://www.iejgeo.com/matder/dosyalar/makale-11/paper2(bayar-ekmekci-ozcan).pdf.